BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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CSE 200: Technical Writing and Presentation

Diffusion Models

An Overview of Progressive Denoising for Image Generation

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Abstract

Diffusion models have become an important class of generative models because they can produce high-quality images without relying on adversarial training. Their basic idea is to gradually corrupt data with Gaussian noise and then learn how to reverse that process. This report explains the forward and reverse diffusion processes, the role of the U-Net architecture, and the standard noise-prediction objective used in Denoising Diffusion Probabilistic Models (DDPMs) [1, 2]. It also briefly discusses faster sampling methods, latent diffusion, common evaluation metrics, and several well-known applications [3, 4]. Finally, the report points out practical limitations such as slow sampling and high computational cost, along with broader concerns related to responsible use.

Contents

1 Introduction **1**

2 The Diffusion Process **1**

2.1 Forward Diffusion Process 2

2.2 Reverse Diffusion Process 2

3 Model Architecture and Training **3**

3.1 Noise-Prediction Objective 3

3.2 Training Workflow 3

3.3 Sampling and Guidance in Practice 4

4 Why Diffusion Models Perform Well **4**

5 How Performance is Evaluated **5**

5.1 Visual and Human Evaluation 5

5.2 Fréchet Inception Distance (FID) 5

5.3 Inception Score (IS) 6

5.4 Precision, Recall, and Fidelity-Diversity Trade-off 6

5.5 Text Alignment in Conditional Models 7

5.6 Computational Efficiency Metrics 7

5.7 Why Multiple Metrics Matter 7

6 Important Practical Variants **8**

6.1 Noise Schedule and Sampling Intuition 8

6.2 Latent Diffusion 9

7 Applications **10**

7.1 Text-to-Image Synthesis 10

7.2 Image Editing and Manipulation 11

7.3 Artistic Rendering and Style Transfer 11

7.4 Video Generation 11

7.5 Medical Imaging Applications 11

7.6 Scientific and Design Applications 11

8 Limitations and Responsible Use **12**

8.1 Slow Sampling 12

8.2 High Computational Cost 12

8.3 System Complexity 12

8.4 Ethical and Social Concerns 13

9 Conclusion **13**

List of Figures

1 Real and generated image comparison 1

2 Diffusion model overview 2

3 Progressive noising example 2

4 Simplified diffusion training diagram 3

5 Evaluation metrics landscape 6

6 Conceptual signal-noise balance over time 9

7 Latent diffusion pipeline 9

8 Representative diffusion-based applications 10

List of Tables

1 Notation summary for diffusion models 4

2 Comparison of common generative model families 5

3 Evaluation metrics summary 8

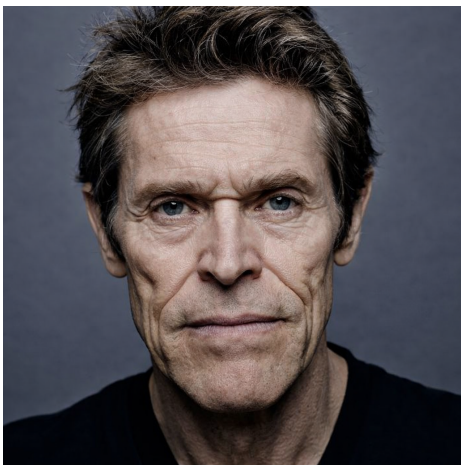
4 Selected practical variants of diffusion models 10

1 Introduction

Generative modeling is concerned with learning the underlying distribution of data so that new samples can be produced from it. For images, this is difficult because realistic output must preserve both fine texture and overall structure. Diffusion models are especially interesting in this setting because the core idea is simple, yet the results are strong in practice [1, 3]. This line of work is also closely related to score-based generative modeling, where noisy samples are used to learn an iterative refinement process [5].

Figure 1 shows why this topic matters. The generated face on the right looks plausible even though it does not belong to a real person. A useful generative model must therefore do more than memorize training examples; it must learn enough about the data distribution to create new samples that still look realistic.

(a) A real face sample.



(b) A model-generated face sample.

Figure 1: Example of a real image and a model-generated image. Good generative models aim to produce new samples that remain visually consistent with real data.

What makes diffusion models intuitive is that they do not try to generate the final image in a single step. During training, a real image is gradually corrupted with Gaussian noise, and the model learns to predict the noise added at different stages. During sampling, the process is reversed: the model starts from random noise and gradually reconstructs structure. This stepwise formulation makes the learning problem easier to handle than direct one-shot image generation.

2 The Diffusion Process

The workflow of a diffusion model has two complementary parts: a *forward diffusion process* that destroys information and a *reverse diffusion process* that reconstructs structure. Figure 2 summarizes this idea in one diagram.

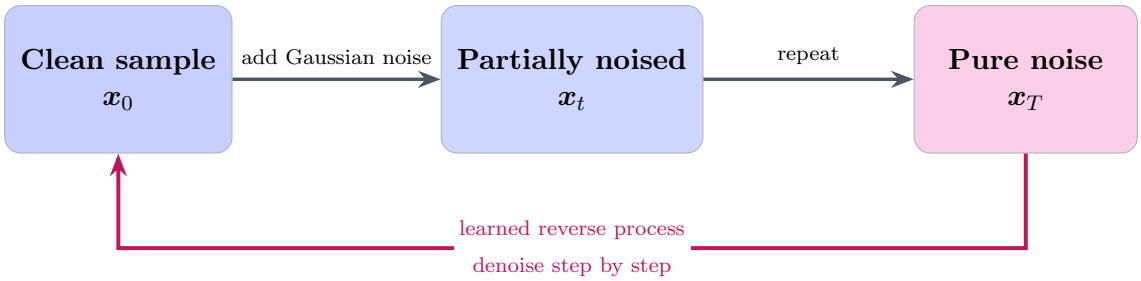


Figure 2: Conceptual overview of diffusion models. Training uses the controlled noising path, while sampling uses the learned reverse path.

2.1 Forward Diffusion Process

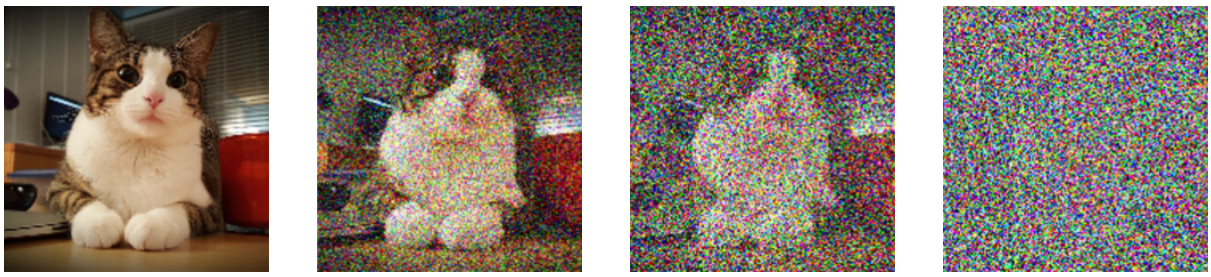
The forward process is a Markov chain that progressively perturbs a clean sample $\boldsymbol{x}_0 \in \mathbb{R}^d$ with Gaussian noise. At each timestep t, a small variance term β_t determines how much corruption is introduced:

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I}\right). \tag{1}$$

If we define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, then a sample at any timestep can be written directly as

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}). \tag{2}$$

Equation 2 is convenient during training because it allows the model to sample a noisy state at any timestep directly, without simulating all previous steps.



(a) Clean (b) Low noise (c) Heavy noise (d) Pure noise

Figure 3: Visual interpretation of the forward diffusion process. A structured image is gradually transformed into an approximately Gaussian sample.

2.2 Reverse Diffusion Process

Once the data has been corrupted, the generative task becomes the reverse problem: infer how to move from $\boldsymbol{x}_t$ back to a less noisy sample $\boldsymbol{x}_{t-1}$. The model parameterized by θ learns a reverse distribution of the form

$$p_\theta(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t)). \tag{3}$$

In practice, many implementations train a neural network to predict the added noise rather than predict clean pixels directly. This reparameterization simplifies learning because the network only needs to estimate the perturbation that separates $\boldsymbol{x}_t$ from the underlying clean sample [1, 2].

3 Model Architecture and Training

Besides the probabilistic formulation, the choice of network architecture also matters. Most image diffusion systems use a U-Net-style backbone with multi-scale feature extraction and skip connections [6]. The network takes two main inputs: a noisy image $\boldsymbol{x}_t$ and an embedding of the timestep t.

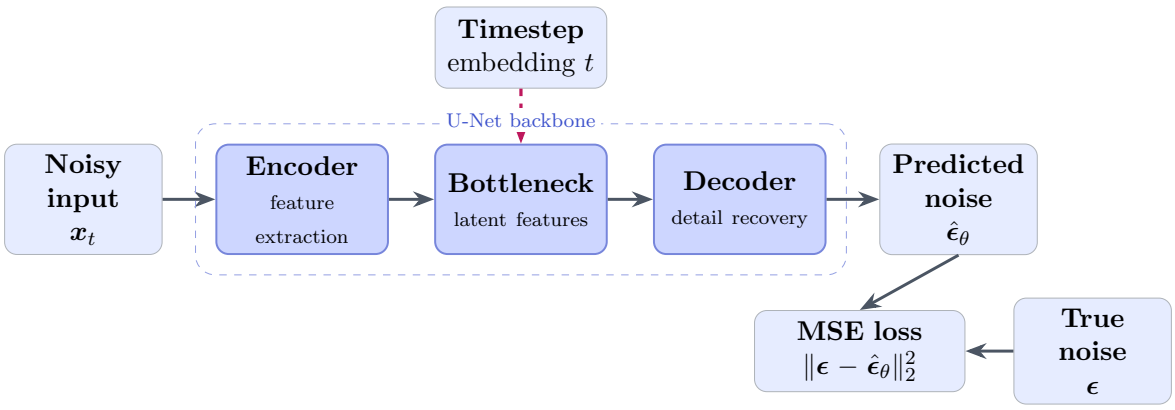


Figure 4: A simplified training view of an image diffusion model. The network receives a noisy image and the current timestep, predicts the noise component, and is trained with a mean-squared error objective.

3.1 Noise-Prediction Objective

The standard DDPM training objective is

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{\boldsymbol{x}_0,t,\boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\|_2^2 \right]. \tag{4}$$

This objective is convenient because the target noise $\boldsymbol{\epsilon}$ is known exactly; it is produced by the training procedure itself. That gives the model a clear supervision signal. In effect, the network is trained to solve many smaller denoising problems instead of one difficult image-generation problem.

3.2 Training Workflow

The complete training routine can be summarized as Algorithm 1. The procedure alternates between sampling a real image, corrupting it to a randomly chosen timestep, and updating the network so that its predicted noise matches the actual perturbation.

Algorithm 1 Training a diffusion model with noise prediction

Require: Training dataset $\mathcal{D}$, noise schedule $\{\beta_t\}_{t=1}^{T}$, neural network $\boldsymbol{\epsilon}_\theta$

```
while training has not converged do
    Sample a clean image x_0 ~ D
    Sample a timestep t uniformly from {1, ..., T}
    Sample Gaussian noise ε ~ N(0, I)
    Form a noisy input using Equation 2
    Predict ε̂ = ε_θ(x_t, t)
    Compute L_simple = ||ε − ε̂||_2^2
    Update θ using gradient descent
end while
```

3.3 Sampling and Guidance in Practice

At inference time, generation starts from $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and repeatedly applies the learned reverse transitions until $\boldsymbol{x}_0$ is obtained. In practice, the main drawback is that this reverse process can be slow. One common way to reduce the number of steps is DDIM sampling [3]. Another practical idea is guidance, which pushes the model to follow a condition more strongly. Examples include classifier guidance and classifier-free guidance [7, 8]. These methods are especially important in modern text-to-image systems because they improve control over the final result.

Table 1 collects the most important symbols used in the mathematical discussion so that the notation remains easy to follow.

Symbol	Description
$\boldsymbol{x}_0$	A clean training sample drawn from the data distribution.
$\boldsymbol{x}_t$	The noisy version of the sample at diffusion step t.
β_t	Variance schedule controlling how much noise is added at step t.
α_t	Shorthand for $1 - \beta_t$.
$\bar{\alpha}_t$	Cumulative product $\prod_{s=1}^{t} \alpha_s$, measuring the remaining signal after t steps.
$\boldsymbol{\epsilon}$	Gaussian noise used to corrupt the sample.
$\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)$	Neural network prediction of the added noise.

Table 1: Notation used in the forward process, reverse process, and training objective.

4 Why Diffusion Models Perform Well

There are several practical reasons diffusion models perform well. First, once Gaussian noise is sampled, the training target is known analytically, which avoids the adversarial instability often seen in GAN training. Second, iterative denoising lets coarse structure appear before fine details are refined. Third, conditioning inputs such as text, class

labels, or masks can be integrated naturally into the denoising network, which makes the framework useful across many tasks.

These characteristics help explain why diffusion systems became competitive with, and in many settings preferable to, earlier generative approaches. Table 2 summarizes typical trade-offs.

Model family	Training stability	Sample quality	Sampling speed	Main comment
VAE	High	Moderate	Fast	Strong probabilistic interpretation but often blurrier outputs.
GAN	Moderate to low	High	Very fast	Can generate sharp images but training may be unstable and prone to mode collapse.
Diffusion model	High	Very high	Slow	Excellent fidelity and diversity, but generation requires many denoising steps.

Table 2: A high-level comparison of common generative model families. The table emphasizes typical tendencies rather than strict universal rules.

5 How Performance is Evaluated

Evaluating diffusion models is not straightforward because there are multiple aspects to consider. Researchers need to judge image quality, check that the model follows instructions correctly, and assess whether the model is actually generating diverse new images or just memorizing training data. To do this comprehensively, multiple evaluation metrics are typically used together.

5.1 Visual and Human Evaluation

The most direct way to assess whether an image looks good is simply to look at it. Papers about diffusion models usually include sample images showing what the model can generate, sometimes arranged in grids to showcase diversity or arranged side-by-side to compare against competing methods. Beyond just showing images, some studies conduct user studies where human raters score generated images on qualities like realism, clarity, and how well they match a given description. While human evaluation is time-consuming and subjective, it remains an important sanity check because humans are ultimately the audience for most generated images.

5.2 Fréchet Inception Distance (FID)

One widely used automatic metric is the Fréchet Inception Distance, or FID. The basic idea is to use a pre-trained image classification network (usually InceptionV3, trained on ImageNet) to extract feature representations from both real images and generated

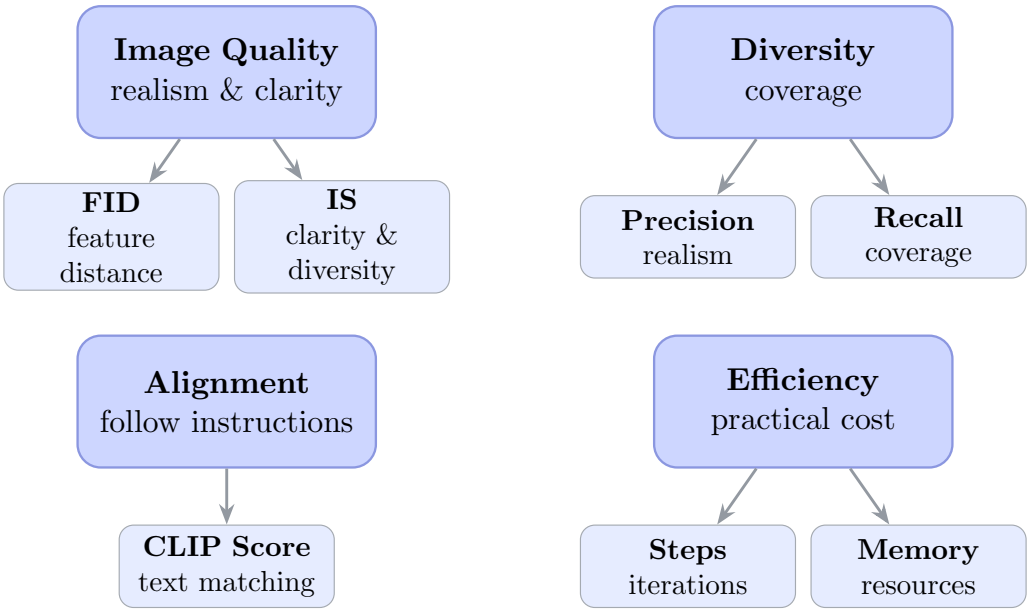


Figure 5: Overview of evaluation metrics organized by the research concerns they address. Most papers report metrics from multiple categories to provide a complete picture.

images. The FID metric then measures the statistical distance between these two sets of features—specifically, it computes the distance between the Gaussian distributions that best fit each set. Lower FID values indicate that generated images are statistically more similar to real images in terms of visual features [9]. FID is popular because it correlates reasonably well with human perception of quality, but it is not perfect; it does not capture every aspect of what makes an image look realistic.

5.3 Inception Score (IS)

Another common metric is the Inception Score, which uses the same pre-trained InceptionV3 network but measures two things: first, whether generated images look clear and recognizable (high prediction confidence), and second, whether the model generates diverse images across different classes (high entropy across predictions). A good generative model should score high on both counts. The main limitation of IS is that it only works well for images of objects and classes that InceptionV3 was trained to recognize; it is less meaningful for artistic or abstract images.

5.4 Precision, Recall, and Fidelity-Diversity Trade-off

To separate the concepts of quality and diversity, researchers often compute precision and recall metrics. Precision measures what fraction of generated images are realistic (avoiding fake-looking outputs), while recall measures what fraction of the training data distribution is actually covered by the generated samples (avoiding mode collapse). These metrics help diagnose whether a model's failure is due to generating poor quality images or failing to explore the full range of possible outputs. In practice, there is often a trade-off: pushing the model to generate higher-quality images sometimes causes it to focus on

safer, more typical examples and miss rare variations.

5.5 Text Alignment in Conditional Models

For text-to-image models, image quality alone is not sufficient because the output must also match the text prompt. Researchers use metrics like CLIP Score, which uses a pre-trained vision-language model (CLIP) to measure how well a generated image aligns with a given text description. A generated image gets a high score if CLIP thinks the image and text are closely related. This metric directly addresses the user's need: a beautiful image that completely ignores the prompt is not useful.

5.6 Computational Efficiency Metrics

Beyond image quality, practical deployment requires considering computational cost. Researchers often report how many denoising steps are needed for a model to produce acceptable results, how long generation takes on standard hardware, and how much GPU memory is consumed. These metrics matter because they determine whether a model can actually be used in real applications. A theoretically perfect model that requires a supercomputer is less useful than a slightly lower-quality model that runs on consumer hardware.

5.7 Why Multiple Metrics Matter

Because no single metric captures everything, research papers typically report several metrics together. A model with high FID but low Inception Score might be generating realistic-looking objects but missing diversity. A model with high CLIP Score but low FID might follow text prompts well but generate images with visual artifacts. By reporting multiple metrics—both automatic and human-based—researchers give readers a more complete picture of what the model can and cannot do well. When comparing two competing systems, differences in individual metrics can sometimes conflict, which is why the research community values both quantitative metrics and qualitative analysis. Ultimately, the right metric depends on the specific application: a model used for creative exploration might prioritize diversity, while a model used for product rendering might prioritize photorealism.

Metric	Measures	Direction	Key limitation
FID	Statistical distance between real and generated features	Lower is better	Only uses pre-trained features
IS	Clarity & diversity combined	Higher is better	Biased toward ImageNet categories
Precision	Realism of generated samples	Higher is better	Doesn't catch mode collapse
Recall	Coverage of training distribution	Higher is better	Doesn't guarantee realism
CLIP Score	Text-image alignment	Higher is better	Ignores visual artifacts
Human Eval	Subjective quality assessment	Context-dependent	Limited sample size

Table 3: Common diffusion model evaluation metrics at a glance.

6 Important Practical Variants

The basic DDPM formulation is enough to understand the core idea, but many real systems use modified samplers or compressed representations to make the method more practical. Two widely discussed examples are DDIM sampling [3] and latent diffusion [4]. These variants do not change the main idea of progressive denoising; instead, they make the framework more usable in real settings.

6.1 Noise Schedule and Sampling Intuition

Although the forward process is defined mathematically, it is helpful to think of diffusion in terms of how signal gradually gives way to noise. Early timesteps preserve most of the original structure, while later timesteps are dominated by noise. This transition is controlled by the schedule $\{\beta_t\}$ and the cumulative factor $\bar{\alpha}_t$.

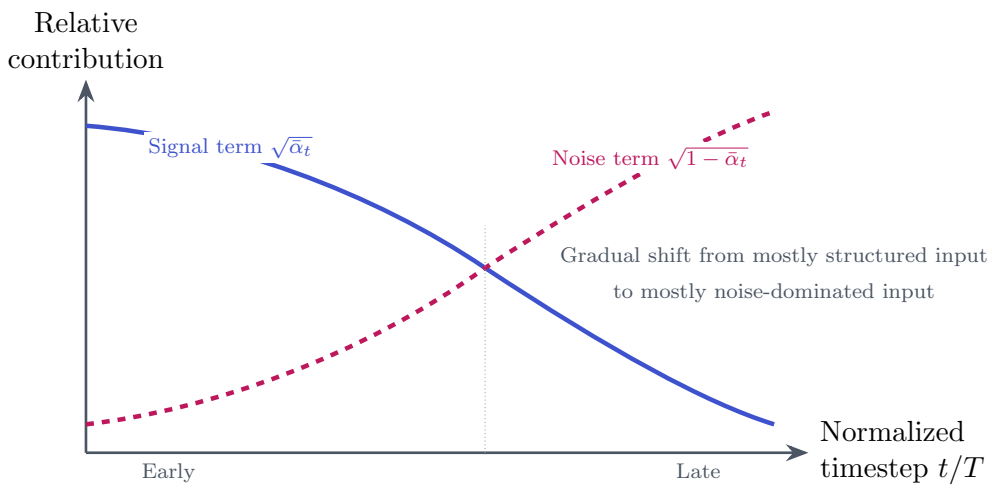


Figure 6: Conceptual signal-noise balance over diffusion timesteps. The plot is schematic and illustrates how the contribution of the original signal decreases while the contribution of noise increases.

Figure 6 is only a conceptual sketch, but it helps explain why sampling can be expensive: the model has to undo a long sequence of small changes. DDIM-style samplers address this issue by following a more direct denoising path, often producing useful samples with fewer steps than a standard DDPM sampler [3].

6.2 Latent Diffusion

Working directly in pixel space is expensive, especially for high-resolution images. Latent diffusion reduces this cost by encoding an image into a lower-dimensional latent representation, running diffusion in that space, and then decoding the result back into pixel space [4]. Because the denoising network operates on a compressed representation, both memory use and sampling cost can be reduced substantially.

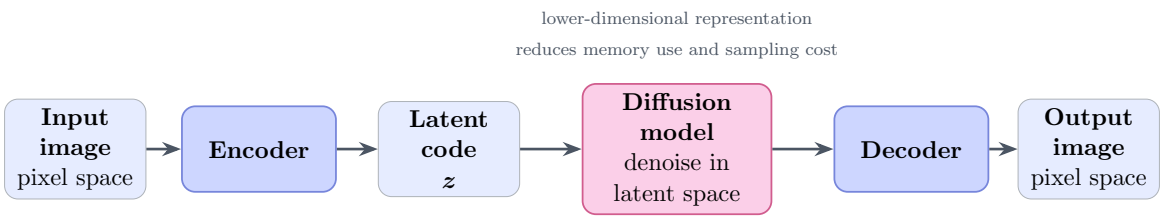


Figure 7: Simplified view of latent diffusion. Instead of applying diffusion directly in pixel space, the model denoises a compressed latent representation and then decodes the result back into an image.

This is one reason modern text-to-image systems can work at fairly high resolutions. The denoising process is still learned in the same way, but the computation is carried out in a more compact space. Table 4 summarizes the practical motivation behind several common variants.

Variant	Main idea	Practical benefit	Typical trade-off
DDPM	Learn the reverse diffusion chain in pixel space.	Strong baseline formulation and high sample quality.	Often requires many denoising steps.
DDIM	Use an implicit sampling path that can skip some steps.	Faster generation with fewer sampling steps.	Speed and quality depend on the chosen schedule.
Latent diffusion	Apply diffusion to a compressed latent representation.	Lower memory cost and more practical high-resolution synthesis.	Requires an additional encoder-decoder stage.

Table 4: Selected practical variants built on the basic diffusion framework.

7 Applications

Diffusion models have moved beyond being a research curiosity and are now integrated into both popular consumer tools and specialized technical workflows. The versatility of the approach means that diffusion-based systems are useful across many different domains, from creative content generation to scientific research.

7.1 Text-to-Image Synthesis

One of the most visible applications of diffusion models is converting text descriptions into images. Systems like DALL-E, Midjourney, and Stable Diffusion allow users to describe what they want to see, and the model generates an image matching that description. This capability has democratized image creation by removing the need for drawing or photography skills. A designer can quickly prototype ideas, a writer can illustrate a concept for a story, or a marketer can generate ad copy images without hiring a photographer. These tools have become popular enough that they are now used in real professional workflows, though they still have limitations in understanding complex or unusual requests.

(a) DALL-E. (b) Midjourney. (c) Sora.

Figure 8: Examples of widely recognized systems associated with diffusion-based generation and editing.

7.2 Image Editing and Manipulation

Beyond creating images from scratch, diffusion models can also be used to edit existing images. Instead of starting from random noise, the model can start from a partially noised version of an input image and then denoise it while respecting certain constraints. This allows for operations like inpainting (filling in missing regions), style transfer (applying one image's characteristics to another), or targeted object removal. These capabilities are beginning to appear in consumer photo editing software, making advanced image manipulation available to non-experts.

7.3 Artistic Rendering and Style Transfer

Diffusion models can be fine-tuned or conditioned to generate images in particular artistic styles. An artist might train a model on their own work or a particular aesthetic and then use it to create variations or explore new directions. Video game studios are exploring diffusion for quickly generating concept art, and entertainment companies use these models to speed up aspects of animation and rendering pipelines. While the results still often need human refinement, diffusion-based tools can help artists iterate faster.

7.4 Video Generation

Video generation is an emerging frontier. Systems like Sora demonstrate that diffusion models can be extended to generate short video clips from text descriptions or static images. Video generation is considerably harder than image generation because the model must maintain temporal coherence—objects should move smoothly, lighting and perspective should make sense across frames—but the same underlying denoising principles apply. As this technology matures, it could change how video content is produced, from special effects to previsualization of scenes.

7.5 Medical Imaging Applications

In healthcare, diffusion models are proving useful for several tasks. They can help denoise noisy medical scans to improve image clarity. They can reconstruct missing or corrupted parts of scans. They can also generate synthetic medical images for data augmentation, which is valuable in medical AI since real patient data is often limited and sensitive. Because medical imaging involves precise technical requirements, this application area requires careful validation, but the potential to improve diagnosis and reduce patient radiation exposure is significant.

7.6 Scientific and Design Applications

Beyond imaging, diffusion models are being explored in more specialized scientific contexts. Researchers are investigating their use for molecule generation in drug discovery, where the model learns to design new chemical structures that might be effective drugs.

In engineering and design, diffusion can be applied to inverse problems—for example, optimizing material shapes or configurations to meet performance targets. These applications require adapting the core framework to work with non-image data, but the principle of iterative refinement remains powerful.

8 Limitations and Responsible Use

Despite their strengths, diffusion models still have several important limitations and concerns that deserve careful attention.

8.1 Slow Sampling

One major practical limitation is that generating a single high-quality image takes a long time. A typical diffusion model requires anywhere from 50 to 1000 denoising steps to produce a good result, and each step involves a full forward pass through the neural network. This means that on standard hardware, generating a single image might take anywhere from several seconds to a minute or more. For comparison, GANs or simpler models can produce images almost instantly. This slowness is a real problem when diffusion models are deployed in applications where users expect quick feedback, such as interactive design tools or mobile applications. Techniques like DDIM sampling help reduce this cost, but the fundamental trade-off between speed and quality remains a practical challenge.

8.2 High Computational Cost

Training a diffusion model from scratch is extremely expensive in terms of computation and energy. State-of-the-art image generation models require thousands of hours of GPU time on expensive hardware like NVIDIA A100 or H100 GPUs. Beyond just the hardware cost, this level of training consumes substantial electrical power, contributing to both financial expense and environmental impact. For researchers and organizations without access to large GPU clusters or cloud computing budgets, training a competitive model from scratch is simply not feasible. This creates a barrier to entry in the field and means that most practical applications rely on either fine-tuning smaller pre-trained models or using existing systems through APIs rather than training from the ground up.

8.3 System Complexity

Building a practical, production-ready diffusion system is more complex than just training the denoising network. In real deployments, many additional components are needed. For instance, text-to-image systems require a text encoder to understand the prompt, a separate module to handle numerical conditioning scores that balance how strictly the model should follow the prompt. Safety filters are often added to prevent generating harmful or inappropriate content. Post-processing pipelines may be needed to handle edge cases or improve the output. All of these components must work together carefully,

and misconfiguring any of them can degrade results. This complexity makes deploying diffusion models in production more challenging than simply using a trained neural network.

8.4 Ethical and Social Concerns

Beyond technical limitations, there are important ethical issues to consider. Diffusion models trained on internet data can easily pick up and amplify societal biases present in their training data. If a training dataset contains stereotypes or skewed representations of certain groups, the model will likely reproduce or even exaggerate those patterns. Additionally, these models can learn to imitate the artistic style of contemporary artists without permission or recognition, raising questions about consent and intellectual property. Finally, diffusion models can be misused to create convincing fake videos or altered images—so-called *deepfakes*—that spread misinformation.

For these reasons, technical progress in diffusion models should be paired with responsible practices: datasets should be curated and documented transparently, usage policies should be clear about what the technology can and cannot be used for, and credit should be given appropriately to the artists and creators whose work influenced the model. Developers and organizations deploying these systems have a responsibility to think carefully about both the intended and potential unintended consequences of their work.

9 Conclusion

Overall, diffusion models show how a difficult generation problem can be broken into a sequence of smaller denoising steps. That is what makes the framework both interesting to study and useful in practice. It brings together probability, optimization, neural networks, and real image-generation systems in a way that is easier to follow than many earlier generative approaches.

At the same time, the method is not free of cost: sampling is slow, training is expensive, and responsible use matters. Even with these limitations, diffusion models remain one of the most influential ideas in modern generative AI.

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